

Saravanakumar A

RTL-Driven VLSI Design Enthusiast

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Summary

Aspiring Electronics and Communication Engineer specializing in VLSI design and Embedded Systems. Proficient in Verilog, System Verilog for RTL design and functional verification. Dedicated to developing power-efficient architectures and real-time stabilization systems. Seeks to apply technical expertise in digital design principles to deliver innovative hardware solutions in a collaborative engineering environment.

Education

B.E - Electronics and Communication Engineering	2022 – Present
Chettinad College of Engineering and Technology, Karur.	CGPA: 7.6
HSC - 12th Grade	2022
Shree Kalaimagal Matric Higher Secondary School, Kulithalai.	Percentage: 71%
SSLC - 10th Grade	2020
Saraswathy Vidhyalaya Matriculation School, Lalapet.	Percentage: 91.8%

Internship Experience

VLSI Intern , TARAS systems & Solutions, Coimbatore.	07/2025 – 08/2025
- Gained hands-on experience in Digital Electronics fundamentals and design flow. - Practiced coding in Verilog for logic design and simulation. - Enhanced programming proficiency in C related to hardware description. - Understood the application of VLSI design principles in real-world projects.	
VLSI Intern , TARAS systems & Solutions, Bangalore.	12/2024– 06/2025
- Completed a 6-month online internship focused on VLSI Design and Verification. - Gained hands-on experience in Verilog, System Verilog, and UVM for RTL design and functional verification. - Improved understanding of the VLSI design flow through structured sessions and project-based learning. - Strengthened skills in digital design principles and simulation methodologies.	

Skills

Technical Skills: Verilog, System Verilog, C, Python, VHDL, Embedded C, RTL Design.
Hardware: Arduino, ESP8266 (NodeMCU), Microcontrollers (8051), MPU6050.
Tools: Xilinx Vivado, MATLAB, EDA Playground, Cisco Packet Tracer, Tinkercad
Soft Skills: Problem Solving, Critical Thinking, Collaboration, Adaptability, Debugging, Leadership.

Projects

AeroGuard - Real-Time Missile-Inspired Stabilization
Tools Used: ESP8266 (NodeMCU), MPU6050 IMU Sensor, SG90 Servo Motors, MATLAB, Embedded C.
Purpose: To implement real-time angle correction (roll, pitch, yaw) using PID control and intelligent fin stabilization dynamics.
End User: Aerospace researchers and autonomous defense monitoring systems.
Low Power High Performance Vedic Multiplier
Tools Used: Xilinx ISE, Verilog.
Purpose: To design a power-efficient architecture that optimizes speed and power consumption for digital processors.
End User: High-speed computing units and Digital Signal Processing (DSP) applications.
Smart Bridge Using Arduino
Tools Used: Arduino, Servo Motor, Soil Moisture Sensor, Embedded C.
Purpose: To develop an automated response system that detects rising water levels to prevent flood-related infrastructure failures.
End User: Civil infrastructure authorities and public safety departments.

Certifications and Achievements

Secured 2nd Prize in Inter-College Mini Project Expo for the "AeroGuard - Real-Time Missile-Inspired Stabilization" project. Secured 3rd Prize in Inter-College Mini Project Expo for the "Low Power High Performance Vedic Multiplier" project Awarded Robotics Starter Kit by AMD for the "AI-powered electric scooter" concept in the Pervasive AI Developer Contest. Completed VLSI Course Level-1 (1-week training) at Taras Solutions, focusing on digital logic and circuit simulation.
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